

GENERAL THETA FOUNDATIONS I: INTRINSIC ACTION GAPS AND NUMERICAL MEMORY

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ABSTRACT. We study the nonuniform clocked atomic-row label width of numerical experiments generated by a finite orthogonal command alphabet. We formulate expansion on the effective action rather than on a presenting group, and remove the dependence of calibration on a choice of irreducible copies. A commutant covariance formula computes an intrinsic isotypic activation magnitude; an optimized query norm gives a canonical error threshold. For arbitrary finite seeds, let R be their invariant span and let σ be the largest minimum unit-orbit dimension among its nontrivial irreducible types. If one type attaining σ has a positive action gap, then at the stated fixed signal and error the minimum hidden width is $\Theta(N^{\sigma/2})$. No expansion is required on the other types. We exhibit a faithful rational alphabet with an effective circle factor, no full group norm gap, and linear numerical width. We compute the minimum-orbit invariant for all compact simple adjoint representations and for classical defining and second exterior power modules, including their minimizing stabilizers. The lower bound retains the actual hidden register; the upper construction prices its branch labels and realizes every numerical response exactly. Epoch, horizon-specific design, tables, arithmetic and exact atomic sampling remain uncharged; this is not a uniform-space theorem.

1. INTRODUCTION

A numerical command experiment asks a finite register to retain enough information to answer one selected observable after a word of known length. The number of labels at a single cut is a static positive factorization problem. Using small factorizations at every cut requires one compatible chain of stochastic updates, and is a different problem. For orthogonal command dynamics, orbit geometry and a conditional-mean entropy budget quantify this difference.

The hypothesis should be a property of the experiment. A presentation by a group with an unused compact factor can change the full regular representation gap without changing a single response probability. Likewise, decomposing an isotypic space into equivalent irreducible copies can change individual seed projections without changing the experiment. We remove both choices: expansion is measured in the actual action, and activation is optimized over isometric intertwiners. The latter optimization has an exact commutant covariance formula for its unweighted magnitude. These changes also permit nonspanning seeds and experiments with nonexpanding, genuinely acting factors.

Fix orthogonal matrices A_a on $V = \mathbb{R}^D$, indexed by a finite alphabet, and a nonempty finite set $\mathcal{X}$ of unit seeds. For a word $w = a_1 \cdots a_N$ let $A_w = A_{a_N} \cdots A_{a_1}$. With a selected coordinate j and a binary answer B , the law is

$$(1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, A_w x \rangle}{2}.$$

The error is the maximum binary total-variation error over all seeds, words and queries. The width $W_{N,\varepsilon}$ is the minimum of the maximum number of available persistent labels in a realizing

Date: September 26, 2026. Revision 41.

2020 Mathematics Subject Classification. Primary 93B15, 22D10; Secondary 94A17, 68Q10, 81P45.

Key words and phrases. Positive realization, compact group, orbit dimension, entropy, hidden memory, quantum automata, numerical simulation.

machine, including the final two-label answer cut. The horizon and epoch, nonuniform row tables, arithmetic and exact atomic sampling are free. A different machine may be used for every N . These are finite positive realizations, not uniform space-bounded algorithms.

Let $H = \overline{\langle A_a \rangle}$ be connected and put $R = \text{span}(H\mathcal{X})$, $r = \dim R$. Its canonical isotypic decomposition has invariant part R_0 and nontrivial types λ . For one irreducible model E_λ of each active type, define

$$s_\lambda = \min_{\|v\|=1} \dim(Hv), \quad \sigma_{\mathcal{X}} = \max_{\lambda} s_\lambda.$$

The maximum is zero if there is no nontrivial type. For an isometric intertwiner $J : E_\lambda \rightarrow R_\lambda$, let $B(J) = \|J^*\|_{\ell_D^\infty \rightarrow \ell^2}$, and define

$$c_\lambda = \max_{x \in \mathcal{X}, J} \frac{\|J^*x\|}{B(J)}.$$

All these quantities belong to the represented experiment and its specified query geometry. For a command probability p , the action gap on E_λ is $g_\lambda = 1 - \|P_\lambda - \Pi_\lambda\|^2$, where P_λ averages the commands on sphere functions and Π_λ is Haar projection onto the invariant functions.

Theorem 1.1 (Intrinsic dominant-action law). *Suppose $0 < \rho \leq \min\{1/10, 1/(8r)\}$. If an active type λ_* attaining $\sigma_{\mathcal{X}}$ has $g_{\lambda_*} > 0$ and $0 \leq \varepsilon < \rho c_{\lambda_*}/2$, then*

$$(2) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_{\mathcal{X}}/2}).$$

No action gap is required on the other types. The lower bound permits arbitrary hidden states and time-dependent stochastic rows. An exact upper realization uses common command rows and a horizon-dependent terminal decoder, with its randomized branch identity charged to the register. If all active vectors are invariant, constant width suffices.

The action gap is identical on sphere functions and on Lebesgue-square-integrable functions on the representation space. It has an exact harmonic-block norm description. A full regular-representation gap of the effective group remains a sufficient certificate, but is no longer required by the headline. Proposition 8.1 makes the extension strict: ten rational orthogonal commands act faithfully on $\mathbb{R}^3 \oplus \mathbb{R}^2$, generate $\text{SO}(3) \times \text{SO}(2)$, and have no full group norm gap, while their numerical width is $\Theta(N)$. The three-dimensional constituent expands; the circle constituent does not.

We also compute rather than merely name the orbit invariant. Theorem 9.1 determines every compact simple adjoint minimum by a single-node root deletion, including all exceptional types and the minimizing centralizers. Theorem 9.3 determines the invariant throughout the second exterior-power fundamental family. In its $\text{SU}(6)$ example the minimizing orbit is $\text{SU}(6)/\text{Sp}(3)$ of dimension fourteen; the decomposable highest-weight orbit has dimension seventeen. This illustrates why projective highest-weight geometry cannot be substituted for the real-vector orbit required by the memory theorem.

Notation	Meaning
H, R	Effective compact matrix group and invariant span of the seeds.
R_λ, E_λ	Canonical isotypic part and one irreducible model.
$s_\lambda, \sigma_{\mathcal{X}}$	Minimum orbit dimension and dominant active minimum.
g_λ, c_λ	Intrinsic action gap and query-calibrated activation.
$K_t, W_{N,\varepsilon}$	Available labels at a cut and optimal whole-run peak.

The analytic foundation is the orbital entropy occupation theorem, whose complete proof is retained. Uniform orbit charts and contracted orbit hulls supply its geometric lower and constructive upper partners. Those results are inherited; the present extensions are the intrinsic action formulation, canonical activation on arbitrary seed spans, dominant-only expansion, and the

resulting faithful nongapped example. Root centralizers, skew congruence normal forms, covering geometry and positive attenuating embeddings are classical tools, not new names for discoveries. The original projective and arithmetic applications retain their separate signal ranges and external assumptions in the appendices. No main result relies on the optional numerical LPS input.

2. THE EXPERIMENT AND ITS POSITIVE REALIZATION

In the general notation below, G may be taken to be the effective matrix group H of Section 3; $U_a = A_a$. The main hypothesis is the constituent action gap, not the full regular gap of an arbitrary presenting group. Older sufficient full-group hypotheses are explicitly retained only where stated. Let G be a compact group, let $U : G \rightarrow O(V)$ be a continuous orthogonal representation on $V = \mathbb{R}^D$, and fix a finite command alphabet $\mathcal{A} \subset G$. A finite seed set $\mathcal{X} \subset \mathbb{S}(V)$ is given initially. For $w = (a_1, \dots, a_N)$ write $U_w = U_{a_N} \cdots U_{a_1}$. Relative to an orthonormal basis $e_1, \dots, e_D$, the prescribed numerical experiment is

$$(3) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\}, \quad j \in [D], \quad 0 < \rho \leq 1/10.$$

All seeds, words and queries are required, not only almost-sure words under one source law. The probabilities are between $(1 - \rho)/2$ and $(1 + \rho)/2$.

Definition 2.1 (Clocked atomic-row realization). A horizon- N machine consists of finite sets $\mathcal{S}_t$ with $|\mathcal{S}_t| = K_t$, a stochastic initialization E_x on $\mathcal{S}_0$, stochastic matrices $T_{t,a} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1}$, and columns $d_j : \mathcal{S}_N \rightarrow [-1, 1]$. Its mean response is

$$(4) \quad E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, U_w x \rangle$ by at most 2ε for every (x, w, j) . Available labels, including padding, count. All rows, including unreachable ones, are normalized. Only the updated label persists. The seed, word prefix, old random choices and any data-dependent protocol information cannot be consulted again unless represented by that label. The external time, horizon, fixed program and exact sampling from its rows are free. The final binary answer may be sampled when j arrives; if retained, it uses a two-label final cut. Write $W_{N,\varepsilon}$ for the minimum of $\max(2, K_0, \dots, K_N)$.

The state alphabets and matrices may be chosen anew for each N , but the command symbols and their experimental meanings are fixed. Nothing requires one autonomous device or one decoder to serve all stopping lengths. No adaptive choice of the command is optimized in this model. A complex unitary representation can be treated by its realification; all inner products in the entropy argument are real.

Fix one seed x and an independent command law at each epoch for the converse. Let $Y_t = U_{a_t} \cdots U_{a_1} x$ and let S_t be the actual machine state under this test law. Define

$$(5) \quad Z_t = \mathbb{E}[Y_t \mid S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2.$$

These random vectors and their laws are analytical devices, not an additional runtime register or oracle.

Lemma 2.2 (Quadratic budget and terminal calibration). *For the fresh command a , put $X_t = U_a Z_t$. Then*

$$(6) \quad Z_{t+1} = \mathbb{E}[X_t \mid S_{t+1}], \quad \Delta_t := \mathbb{E}\|X_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The sequences q_t, v_t are nonincreasing and start at one. A machine with error at most ε satisfies

$$(7) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon \quad (0 \leq t \leq N).$$

Proof. The current command is independent of the past, and the new label is generated from (S_t, a) by a common row. Conditioning first on (S_t, a, S_{t+1}) and then on S_{t+1} proves the first identity. The orthogonal projection identity for conditional expectation gives the second; Jensen and orthogonality give both monotonicities. The fixed seed has unit norm, even with randomized initialization. For calibration form $d = (d_1, \dots, d_D)$ only in the proof, so $\|d\| \leq \sqrt{D}$. Conditioning on the complete word and using (3)–(4) gives $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\sqrt{D}\varepsilon$. On conditioning instead on S_N , this is at most $\sqrt{D}q_N$. This proves (7). It does not require a joint measurement of the D coordinates. $\square$

For each fixed profile, the optimum actual error is attained: all stochastic rows and terminal columns form a finite product of compact sets and the objective is a maximum of finitely many continuous functions. Thus the subsequent lower bounds concern the actual optimization, not a sum of local error certificates.

3. EXPANSION OF THE EFFECTIVE ACTION

The stochastic experiment specifies matrices, not a presentation of a group. Let $\mathcal{A}$ be its finite command alphabet, let $A_a \in O(V)$ be its command matrices, and put

$$H = \overline{\langle A_a : a \in \mathcal{A} \rangle} \subset O(V).$$

We assume that this compact Lie group is connected. A probability p on $\mathcal{A}$ is used only in a converse; the required numerical law remains wordwise. All the quantities below are defined from these matrices and p . For a nonzero invariant subspace E , write σ_E for uniform sphere probability and define operators on $L^2(\mathbb{S}(E), \sigma_E)$ by

$$P_E f(u) = \sum_a p(a) f(A_a^{-1} u), \quad \Pi_E f(u) = \int_H f(h^{-1} u) dm_H(h).$$

The squared *action gap* is

$$(8) \quad g_E = 1 - \|P_E - \Pi_E\|_{2 \rightarrow 2}^2.$$

Thus $0 \leq g_E \leq 1$. When every function is invariant, the norm in this formula is zero. Such a subspace has no positive-dimensional orbit and is not used in an occupation lower bound.

Proposition 3.1 (Presentation invariance and the relevant spectrum). *The action gap has the following properties.*

- (1) *It is unchanged by an orthogonal intertwining of the action, by splitting a command into identical letters with the same total mass, or by replacing a presenting group by any compact extension with the same matrices and pushed-forward command law.*
- (2) *On $L^2(E, dz)$, with the same formulas for P_E and Π_E , the optimal inequality is*

$$(9) \quad \|P_E f\|_2^2 \leq \|\Pi_E f\|_2^2 + (1 - g_E) \|f - \Pi_E f\|_2^2.$$

In particular, the sphere and Lebesgue action gaps are identical.

- (3) *Let $\mathcal{H}_l(E)$ be the restrictions of homogeneous harmonic polynomials of degree l , and let $P_{E,l}, \Pi_{E,l}$ be the corresponding finite-dimensional restrictions. Then*

$$(10) \quad 1 - g_E = \sup_{l \geq 0} \|P_{E,l} - \Pi_{E,l}\|_{2 \rightarrow 2}^2.$$

No finite truncation of this supremum is an asserted gap certificate.

- (4) *A mean-zero regular-representation norm bound $\sqrt{1 - g}$ on H implies $g_E \geq g$. It is a sufficient certificate, not the definition of the hypothesis.*

Proof. The operators are determined by the probability measure $\sum_a p(a)\delta_{A_a|E}$ and the effective compact image on E . An orthogonal intertwiner induces a unitary conjugacy of both operators. Haar probability pushes forward to Haar probability on a compact image. This proves the first assertion, including extensions with a nontrivial kernel.

Polar coordinates identify $L^2(E, dz)$ with $L^2((0, \infty), r^{d-1}dr) \hat{\otimes} L^2(\mathbb{S}(E), \sigma_E)$, up to an immaterial fixed normalization. Both operators act as the identity on the radial factor. Consequently $\|P_E - \Pi_E\|$ is the same in the two spaces. Haar invariance gives $P_E \Pi_E = \Pi_E P_E = \Pi_E$, so the invariant and orthogonal components separate, proving (9). The usual orthogonal harmonic decomposition of sphere functions is invariant under every orthogonal matrix; the norm of a block-diagonal operator is the supremum of its block norms. This gives (10).

For the last assertion apply the group norm inequality, for fixed u , to $F_u(h) = f(h^{-1}u)$ and integrate in u . Orthogonal invariance of sphere measure identifies the terms with the corresponding action norms. One may first take continuous f , and then use density. This is the group-to-action transference argument, not an implication from a gap in the defining representation alone. $\square$

Corollary 3.2 (Constituent action hypotheses suffice). *In the orbital occupation theorem below, the full presenting-group assumption can be replaced by $g_E > 0$ on the particular irreducible space in which the conditional centroid is tracked. Its conclusion is*

$$g_E \# \{0 \leq t < N : K_t \leq k\} \leq B_{d,A,s} \kappa^{-3-2/s} k^{2/s}.$$

Other invariant summands need not have a positive action gap.

Proof. Equation (9) is exactly the analytic inequality used in entropy production. Apply Theorem 5.5 in E . The register is still the actual whole-machine register; no constituent identifier is supplied to it. The orbit estimate is that of Theorem 4.1 for the image on E . $\square$

The distinction is substantive even after removing trivially acting factors. Section 8 gives a faithful finite-alphabet experiment with an effective circle factor and no full group norm gap, yet with a matched numerical width law supplied by one dominant expanding constituent. The older full-group results remain valid as corollaries whenever their stronger hypotheses hold.

4. THE SMALLEST ORBIT CONTROLS UNIFORM CONCENTRATION

In this section G is a compact connected Lie group, $U : G \rightarrow O(V)$, and $V^G = \{0\}$. Put

$$(11) \quad s(U) = \min_{u \in \mathbb{S}(V)} \dim(Gu).$$

The minimum is over *all* directions. It can be smaller than the dimension of a generic orbit. Since G is connected, $s(U) \geq 1$.

Theorem 4.1 (Uniform concentration across orbit types). *There is a finite constant $A_U \geq 1$ such that*

$$(12) \quad \sup_{u,v \in \mathbb{S}(V)} m_G\{g : \|U_g u - v\| \leq r\} \leq A_U r^{s(U)} \quad (0 < r \leq 1).$$

This exponent is optimal among uniform power upper bounds: for a unit u_0 with $\dim(Gu_0) = s(U)$ there are $c, r_0 > 0$ such that

$$(13) \quad m_G\{g : \|U_g u_0 - u_0\| \leq r\} \geq c r^{s(U)} \quad (0 < r < r_0).$$

No regular-orbit, constant-stabilizer, or simple-spectrum hypothesis is needed in (12).

Proof. Write $m = \dim G$, $s = s(U)$ and let $\mathfrak{g}$ be the Lie algebra. For $u \in \mathbb{S}(V)$ the differential

$$A_u : \mathfrak{g} \longrightarrow V, \quad X \longmapsto dU(X)u$$

has rank $\dim(Gu) \geq s$. Its s th singular value is continuous in u and is strictly positive on the compact sphere. Its minimum is therefore positive. We give a chart argument that also records why variation of the stabilizer causes no loss in the exponent.

Fix $(g_0, u_0) \in G \times \mathbb{S}(V)$. In a smooth coordinate chart $g = \psi(z, w)$, where $z \in \mathbb{R}^s$ and $w \in \mathbb{R}^{m-s}$, select s coordinate directions and a linear projection $L : V \rightarrow \mathbb{R}^s$ for which

$$D_z[LU_{\psi(z,w)}u]_{(g_0, u_0)} = M$$

is invertible. This is possible by the rank bound. By shrinking to a product of bounded open rectangles $Z \times W$ and a neighborhood J of u_0 , arrange

$$\|D_z[LU_{\psi(z,w)}u] - M\|_{\text{op}} \leq \frac{1}{2}\sigma_{\min}(M) \quad (z, w, u) \in Z \times W \times J.$$

Take Z convex. For fixed (w, u) , integration along the segment from z to z' gives

$$(14) \quad \|LU_{\psi(z,w)}u - LU_{\psi(z',w)}u\| \geq \frac{1}{2}\sigma_{\min}(M)\|z - z'\|.$$

If both points lie in the inverse image of the radius- r ball around v , their projected distance is at most $2\|L\|r$. This inverse image, on the fixed w slice, consequently has diameter at most $4\|L\|r/\sigma_{\min}(M)$, and hence s -dimensional Lebesgue measure at most a constant times r^s . Haar measure has a bounded smooth density in the chosen relatively compact chart. Integration in the remaining $m - s$ coordinates yields the same bound there.

Choose finitely many of these product neighborhoods covering $G \times \mathbb{S}(V)$. For each fixed u , those whose J contains u cover G ; summing the preceding bounds proves (12). All constants are independent of the center v . An empty inverse image contributes zero. This use of a finite cover, rather than a separate density estimate on every orbit, is what controls neighborhoods of singular orbit types.

For the converse, $Gu_0 \simeq G/G_{u_0}$ is a compact smooth embedded s -dimensional submanifold. The pushforward of Haar probability is its normalized invariant Riemannian volume: the induced metric is G -invariant and the invariant probability on this homogeneous space is unique. A smooth local graph at u_0 has volume comparable to r^s in a small ambient ball. This proves (13). If a uniform estimate with exponent $s' > s$ held, comparison at u_0 as $r \downarrow 0$ would contradict it. $\square$

The rank computation in (11) is intrinsic to the representation. For example, given algebraic skew matrices $B_1, \dots, B_m$ spanning its infinitesimal action, the condition $s(U) \leq r$ is the existence of a real vector u with $\sum u_i^2 = 1$ and all $(r+1)$ -minors of $[B_1u \cdots B_mu]$ zero. Thus it is an explicit real-algebraic decision problem for fully specified algebraic data; quantifier elimination applies [22]. This is not a test for a spectral gap, and no efficient algorithm in a succinct representation is asserted.

Corollary 4.2 (Intrinsic occupation exponent). *Suppose the representation satisfies (17). In the setting of Lemma 2.2, $q_N \geq \kappa > 0$ implies*

$$(15) \quad \sum_{\substack{0 \leq t \leq N \\ \bar{K}_t \leq k}} g_t \leq 2048D(16A_U)^{2/s(U)} \kappa^{-3-2/s(U)} k^{2/s(U)}.$$

The estimate allows arbitrary hidden continuations and imposes no bound on the other cuts.

Proof. Apply Theorem 5.5 with the uniform estimate (12). In particular the conditional means may change orbit type at every step; their directions still satisfy the same estimate. $\square$

5. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(16) \quad \mathcal{U}_a f(z) = f(U_a^{-1}z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1}z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(17) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma 5.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (17) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1}z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1}z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (17). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

5.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma 5.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(18) \quad h(X + \tau G_D) - h(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(19) \quad \nabla^2 \log f(y) = -\tau^{-2} I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2} I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2} I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \| f)$ gives $h(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = h(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (18). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 14]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma 5.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(20) \quad h(P_t f) - h(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $\mathbf{h}(h) - \mathbf{h}(f) = \sum_a p_t(a) D(f_a \| h)$. For probability densities u, v , $D(u \| v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (17) and $\|\sqrt{f}\|_2 = 1$. □

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

5.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(21) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma 5.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(22) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi\sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (21) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi\sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. □

Theorem 5.5 (Orbital occupation budget). *Under (17) and (21), let a finite hidden register track a unit-vector command process as in (5). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(23) \quad \sum_{\substack{0 \leq t < N \\ \bar{K}_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048D(16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (22) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas 5.2 and 5.3,

$$\mathbf{h}(f_{t+1}) - \mathbf{h}(f_t) \geq g_t \|\sqrt{f_t} - \Pi\sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma 5.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most

$1 + D\tau^2$. The Gaussian maximum-entropy bound and (6) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (22). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary 5.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(24) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (7) with (23) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

6. EXACT SYNTHESIS FROM A MINIMAL ORBIT

Assume in this section that V is a nontrivial real irreducible orthogonal representation, of dimension d . Choose a unit vector v_0 with orbit dimension $s = s(U)$, and set

$$\mathcal{O} = Gv_0, \quad C = \text{conv } \mathcal{O}.$$

The orbit may be a singular orbit in the unit sphere. As a homogeneous space in its own right it is nevertheless a smooth compact manifold.

Lemma 6.1 (Inradius and quadratic support approximation). *One has $d^{-1}\mathbb{B}(V) \subseteq C \subseteq \mathbb{B}(V)$. There are $c, C_0, h_0 > 0$, depending on U and v_0 , such that for $0 < h \leq h_0$ a set $\mathcal{V}_h \subset \mathcal{O}$ with $|\mathcal{V}_h| \leq C_0 h^{-s}$ satisfies*

$$(25) \quad (1 - ch^2)C \subseteq \text{conv } \mathcal{V}_h \subseteq C \quad \text{whenever } ch^2 < 1.$$

Proof. The mean of $U_g v_0$ is an invariant vector and is zero. Its covariance is I/d : the covariance is self-adjoint and commutes with the representation, so its eigenspaces are invariant; irreducibility and trace one give the claim. This is the real Schur orthogonality calculation, for example [21, Lemma 5]. For unit H , let $f(g) = \langle H, U_g v_0 \rangle$ and $M = \max_g f(g)$. Since $-1 \leq f \leq M$,

$$0 \leq \mathbb{E}[(M - f)(f + 1)] = M - \mathbb{E}f^2 = M - 1/d.$$

Every support function of C is thus at least $1/d$, which proves the inradius statement.

Equip $\mathcal{O}$ with its induced Riemannian metric. Finitely many bounded smooth coordinate patches, with their inverses uniformly Lipschitz on smaller patches, give an intrinsic h -net with at most $C_0 h^{-s}$ points. At a maximizer y of $z \mapsto \langle H, z \rangle$ on $\mathcal{O}$, its tangential first derivative is zero. The second fundamental form of the compact embedded orbit is bounded. Along a unit-speed minimizing geodesic of length at most h , Taylor's formula therefore gives

$$\langle H, y \rangle - \langle H, v \rangle \leq L \|H\| h^2$$

for a nearby net point v , with L independent of H, y . The inradius bound gives $\|H\| \leq dh_C(H)$. Hence $h_{\text{conv } \mathcal{V}_h}(H) \geq (1 - Ldh^2)h_C(H)$ for every H . Taking $c \geq Ld$, and using the support-function criterion, proves (25). $\square$

The covering exponent s and quadratic loss at a support maximum are geometric facts, not new quantization exponents. What is needed for realization is an enclosure compatible with *every command* and *every layer*; the following construction supplies it.

Theorem 6.2 (A minimal-orbit positive realization). *For any finite command alphabet, any finite unit seed set, and $0 < \rho \leq 1/(8d)$, the experiment (3) has an exact realization with at most $C_U N^{s(U)/2}$ available labels, for $N \geq 1$. All command matrices can be chosen independent of the epoch; the final decoder may depend on N . Each row has at most $d + 1$ successors. If the seeds belong to $C \cap \mathbb{S}(V)$, the amplitude range improves to $0 < \rho \leq 1/10$.*

Proof. Choose

$$h_N = \min\{h_0, (8cN)^{-1/2}\}, \quad \eta = 1 - ch_N^2 \geq 1 - 1/(8N), \quad a_t = \eta^{-t}/4.$$

Let $v_1, \dots, v_M$ be the net points. The represented vector of a label i at layer t is $a_t v_i$. These vectors describe the fixed rows; they are not stored as real-valued runtime data.

The polytope $\text{conv}\{v_i\}$ contains ηC and hence a ball of radius η/d . For a unit seed x , $4\rho x \in \eta C$ if $\rho \leq 1/(8d)$, because $4\rho \leq 1/(2d) < \eta/d$. Choose its convex coordinates as the initial distribution. More generally, if $x \in C$ and $4\rho \leq 2/5 < \eta$, the same initialization is valid. For each command a and label i , invariance of C gives

$$\eta U_a v_i \in \eta C \subseteq \text{conv}\{v_j\}.$$

Choose any convex coordinates of this point for the row $T_a(i, \cdot)$. They do not depend on t . Since $a_{t+1}\eta = a_t$, conditional expectation updates the represented vector by exactly U_a . Induction gives mean $\rho U_w x$ after each complete prefix.

Use the terminal column $d_j(i) = a_N \langle e_j, v_i \rangle$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $|d_j(i)| \leq 2/7 < 1$. This proves every numerical conditional law exactly. Carathéodory's theorem gives $d + 1$ successors. Finally $M \leq C_0 h_N^{-s} \leq C_U N^{s/2}$, with the fixed two-state output cut absorbed into the constant. Neither convex feasibility nor row lookup is priced as computation in this model. $\square$

Corollary 6.3 (The irreducible width law). *Suppose a probability law on the finite command alphabet has a full mean-zero group norm gap. For a nontrivial real irreducible V , any nonempty finite unit seed set, fixed $0 < \rho \leq 1/(8d)$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$,*

$$(26) \quad W_{N,\varepsilon} = \Theta_{U,p,\rho,\varepsilon}(N^{s(U)/2}).$$

The same order holds for exact vector-state realizations and for fixed-length realizations with common command rows. For seeds on a minimal orbit the amplitude range is $0 < \rho \leq 1/10$.

Proof. A full group norm gap transfers to (17) with a fixed $g > 0$. Apply Corollary 4.2 at the peak k , using $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. It gives $gN \leq B\kappa^{-3-2/s}k^{2/s}$ and therefore the lower order. The exact machine in Theorem 6.2 is an upper bound for every allowed error, is vector-valued in its representation, and has common command rows. These subclasses lie inside the unrestricted clocked class, so the lower bound applies to them too. The last assertion uses the stronger initialization in that theorem. $\square$

7. CANONICAL ACTIVATION AND THE REACHABLE EXPERIMENT

No spanning hypothesis is needed on the given seeds. Their invariant reachable space is

$$(27) \quad R_{\mathcal{X}} = \text{span}\{hx : h \in H, x \in \mathcal{X}\}, \quad r = \dim R_{\mathcal{X}}.$$

Orthogonality makes its orthogonal complement invariant. Every response in (3) is unchanged when its query vector is projected to $R_{\mathcal{X}}$. Choose the *canonical isotypic decomposition*

$$R_{\mathcal{X}} = R_0 \oplus \bigoplus_{\lambda \in \Lambda} R_{\lambda}, \quad R_0 = R_{\mathcal{X}}^H,$$

where R_λ contains all copies of the nontrivial real irreducible type E_λ . Let $d_\lambda = \dim E_\lambda$, let m_λ be its real irreducible multiplicity, and put

$$s_\lambda = \min_{\|u\|=1, u \in E_\lambda} \dim(Hu), \quad \sigma_\lambda = \max_{\lambda \in \Lambda} s_\lambda.$$

For an empty set Λ we set $\sigma_\lambda = 0$. An isometric intertwiner $J : E_\lambda \rightarrow R_\lambda$ identifies a single irreducible copy, without choosing a decomposition into copies. The set $\mathcal{J}_\lambda$ of such intertwiners is nonempty and compact: it is the intersection of the linear intertwining equations with $J^*J = I$.

Definition 7.1 (Canonical seed and query calibration). Let P_λ be the isotypic orthogonal projection, and define

$$(28) \quad a_\lambda = \max_{x \in \mathcal{X}, J \in \mathcal{J}_\lambda} \|J^*x\|,$$

$$(29) \quad B(J) = \|J^*\|_{\ell_D^\infty \rightarrow \ell_{d_\lambda}^2} = \max_{\|u\|=1} \|Ju\|_1, \quad c_\lambda = \max_{x, J} \frac{\|J^*x\|}{B(J)}.$$

The ℓ^1 and ℓ^∞ norms refer to the actual coordinate queries e_j . Thus c_λ is invariant under changes of model coordinates when the queries are transformed with them, but a physical change of the query family is not declared costless. One always has $a_\lambda/\sqrt{D} \leq c_\lambda \leq a_\lambda$.

The equality defining B is norm duality. Since J is an isometry, $1 \leq B(J) \leq \sqrt{D}$, so maxima exist and denominators are nonzero.

Proposition 7.2 (A spectral formula in the commutant). *For $x \in \mathcal{X}$ set*

$$C_{\lambda,x} = \int_H (hP_\lambda x)(hP_\lambda x)^* dm_H(h) \quad \text{on } R_\lambda.$$

Then

$$(30) \quad a_\lambda^2 = d_\lambda \max_{x \in \mathcal{X}} \lambda_{\max}(C_{\lambda,x}), \quad \frac{\max_x \|P_\lambda x\|}{\sqrt{m_\lambda}} \leq a_\lambda \leq \max_x \|P_\lambda x\|.$$

Every active isotypic type has $a_\lambda > 0$. Both a_λ and c_λ are independent of a splitting into equivalent copies. For each fixed seed, $C_{\lambda,x}$ is the Hilbert–Schmidt orthogonal projection of $(P_\lambda x)(P_\lambda x)^$ onto the matrix commutant. Consequently the spectral formula is finite linear algebra once the isotypic projection is supplied.*

Proof. Haar averaging is a self-adjoint idempotent on the space of real matrices, with image the matrices commuting with H . This proves the last assertion. For an isometric intertwiner J , the self-adjoint operator $J^*C_{\lambda,x}J$ commutes with the irreducible action on E_λ . Its eigenspaces are invariant, so it is scalar, even in complex and quaternionic commutant types. Its trace is $\|J^*x\|^2$; therefore it equals $\|J^*x\|^2 I/d_\lambda$. This gives the upper bound by $\lambda_{\max}(C_{\lambda,x})$.

Conversely, the top eigenspace of $C_{\lambda,x}$ is invariant and contains an irreducible copy of type λ . Choose J with image in that eigenspace. The preceding scalar identity gives equality. The trace is $\|P_\lambda x\|^2$, the ambient dimension is $m_\lambda d_\lambda$, and the top eigenspace has dimension at least d_λ . These facts give both norm bounds in (30). If $P_\lambda x = 0$ for every seed, then it is zero on their invariant span, contrary to the presence of R_λ . The remaining assertions follow from the definitions. $\square$

This formula is not an appeal to a preferred basis of a multiplicity space. For instance, on two identical copies of a real-type d -dimensional irreducible, a seed $(v, v)/\sqrt{2}$ has projection norm $1/\sqrt{2}$ on each displayed copy and norm one on the diagonal copy. Its canonical a_λ is one. Taking a minimum over the displayed copies would produce a coordinate-dependent calibration. This seed need not span the ambient space; its reachable space is precisely the diagonal copy.

Theorem 7.3 (Intrinsic numerical width with active seeds). *Assume H is connected, $\mathcal{X}$ is a nonempty finite unit seed set, and $0 < \rho \leq \min\{1/10, 1/(8r)\}$. If $\Lambda \neq \emptyset$, suppose there is a type λ_* with*

$$(31) \quad s_{\lambda_*} = \sigma_{\mathcal{X}}, \quad g_{E_{\lambda_*}} > 0, \quad 0 \leq \varepsilon < \rho c_{\lambda_*}/2.$$

Then

$$(32) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_{\mathcal{X}}/2}).$$

No action gap is required on the other types. The upper realization is exact and uses common command rows and an N -dependent decoder. If Λ is empty, an exact constant-width realization exists. All constants are for the fixed experiment, signal and error.

Proof. Fix a type with $g_{E_{\lambda}} > 0$ and choose x, J attaining c_{λ} , writing $a = \|J^*x\| > 0$ and $B = B(J)$. Track $Y_t = U_{w_t}|_{E_{\lambda}} J^*x/a$ under independent commands with law p , conditioned on the *actual* register of any candidate machine. For its decoder vector $d(S_N) \in [-1, 1]^D$, the wordwise error implies

$$\mathbb{E}\langle JY_N, d(S_N) \rangle \geq \rho a - 2\varepsilon B.$$

Indeed JY_N lies in the chosen invariant copy, its pairing with $U_w x$ is a , and its ℓ^1 norm is at most B for each word. On the other hand $\|J^*d(S_N)\| \leq B$, so conditioning on S_N bounds this correlation above by Bq_N . Thus

$$(33) \quad q_N \geq \kappa_{\lambda} := \rho c_{\lambda} - 2\varepsilon > 0.$$

Corollary 3.2 and the all-direction orbit bound now give $W_{N,\varepsilon} \geq cN^{s_{\lambda}/2}$. Apply this to λ_* . The proof charges no free information about J or the isotypic type: these are analytical choices for a converse.

For the upper bound choose, solely to construct a machine, any orthogonal irreducible decomposition of $R_{\mathcal{X}}$. Give a copy of dimension d_i the input-independent branch weight $w_i = d_i/r$. In that branch initialize the minimal-orbit construction of Theorem 6.2 at mean $(\rho/w_i)\pi_i x$. Its norm is at most $\rho r/d_i \leq 1/(8d_i)$, the sufficient initialization radius there. The same proof works for subunit initial vectors. Query j uses the restriction of the ambient functional e_j , of norm at most one. The disjoint union of branch labels stores the selected branch. Its weighted output mean is exactly $\rho U_w x$. A fixed subspace is implemented by the scaled cross-polytope construction in the proof of Theorem A.1; give it weight $\dim R_0/r$. The resulting total label count is

$$O(1) + \sum_{\lambda} m_{\lambda} C_{\lambda} N^{s_{\lambda}/2} = O(N^{\sigma_{\mathcal{X}}/2}).$$

All copies are fixed in number. Command rows are common in time; branch selection and the final two-label output cut are charged. If only fixed vectors are reachable, retain the finite seed identity or use the fixed polytope, and keep its state unchanged. $\square$

Corollary 7.4 (Canonical occupation and actual error). *For each type with $g_{E_{\lambda}} > 0$, set $B_{\lambda} = B_{d_{\lambda}, A_{\lambda}, s_{\lambda}}$ from the orbit theorem. Every machine of error ε with $\kappa_{\lambda} = \rho c_{\lambda} - 2\varepsilon > 0$ obeys*

$$g_{E_{\lambda}} \# \{t < N : K_t \leq k\} \leq B_{\lambda} \kappa_{\lambda}^{-3-2/s_{\lambda}} k^{2/s_{\lambda}}.$$

For a prescribed profile of peak K , the optimum actual error is at least

$$(34) \quad \frac{1}{2} \max_{\lambda: g_{E_{\lambda}} > 0} \left[\rho c_{\lambda} - \left(\frac{B_{\lambda} K^{2/s_{\lambda}}}{g_{E_{\lambda}} N} \right)^{1/(3+2/s_{\lambda})} \right]_+.$$

An empty maximum means zero. In particular only one calibrated expanding type of maximal s_{λ} is needed for the matched exponent.

Proof. Use (33) in the same occupation theorem, then solve its peak form for ε and maximize the resulting lower bounds. If the actual error exceeds a type's threshold, its displayed bound is already true. Compactness of all rows and decoders for a fixed profile gives the same inequality for the optimum. $\square$

Proof of Theorem 1.1. Apply Theorem 7.3 and Proposition 3.1. The definitions use only the effective command action, its reachable seed space, and the actual query norms. $\square$

8. A FAITHFUL ACTION WITHOUT A FULL GROUP GAP

The action formulation is not merely a way to remove a superfluous factor from the notation. Here the nonexpanding factor acts nontrivially and is observed by the experiment.

Proposition 8.1 (An expanding dominant summand is enough). *Let $R_x, R_z \in \mathrm{SO}(3)$ be rotations about the first and third axes with cosine $3/5$ and sine $4/5$, and let*

$$\mu_3 = \frac{1}{5}(\delta_I + \delta_{R_x} + \delta_{R_x^{-1}} + \delta_{R_z} + \delta_{R_z^{-1}}).$$

Let $R \in \mathrm{SO}(2)$ have the same cosine and sine, and let $\nu = \frac{1}{2}(\delta_I + \delta_R)$. The ten block-diagonal commands with product law $\mu_3 \otimes \nu$ act faithfully on $V = \mathbb{R}^3 \oplus \mathbb{R}^2$. For seeds $\{\pm e_1, \dots, \pm e_5\}$, $0 < \rho \leq 1/40$ and $0 \leq \varepsilon < \rho/(2\sqrt{3})$,

$$(35) \quad W_{N,\varepsilon} = \Theta(N).$$

The effective group is $\mathrm{SO}(3) \times \mathrm{SO}(2)$ and its mean-zero regular convolution norm is one. Thus (35) is outside the full-effective-group-gap hypothesis as well as outside the full presenting-group-gap hypothesis.

Proof. The complex number $(3 + 4i)/5$ is not a root of unity: a root of unity and its inverse are algebraic integers, whereas their sum $6/5$ is rational and not integral. The closure of its powers is therefore the circle. Powers of R_x and R_z give the two continuous axis rotation subgroups. Their infinitesimal generators and bracket span $\mathfrak{so}(3)$, so they generate a dense subgroup of $\mathrm{SO}(3)$. The product alphabet includes the identities needed to generate the two factors separately, proving the assertion about the effective group.

The matrices are rational. Benoist–de Saxcé’s algebraic-generator theorem [2, Theorem 1.2] supplies a regular spectral gap on the compact simple factor $\mathrm{SO}(3)$. Since μ_3 is symmetric, its operator is self-adjoint and spectral radius equals norm. (Its identity mass also bounds the negative spectral edge away from -1 .) Proposition 3.1 gives $g_{\mathbb{R}^3} > 0$. On the effective circle, put $R = e^{2\pi i \alpha}$ in complex notation. The functions $e^{2\pi i l \theta}$ have eigenvalues $(1 + e^{2\pi i l \alpha})/2$. Infinitely many l have $\|l\alpha\| \rightarrow 0$, by the pigeonhole approximation argument. Their moduli tend to one, proving the failure of a full product norm gap.

The two real irreducible summands have minimum orbit dimensions two and one. The first is dominant and has $c = 1/\sqrt{3}$: its standard basis seeds have projection norm one and its coordinate-query norm is $\sqrt{3}$. Theorem 7.3 now applies with $r = 5$. The exact upper compiler uses $O(N)$ sphere labels and $O(N^{1/2})$ circle labels. No Diophantine assumption on α and no numerical value of the qualitative $\mathrm{SO}(3)$ gap is used. $\square$

The external input in this example is the algebraic spectral-gap theorem for the simple factor, not a newly proved expansion result. The conclusion shows why requiring expansion on every activated type would also be unnecessarily restrictive for a maximum-exponent law.

9. MINIMUM ORBITS FROM ROOT AND SINGULAR-VALUE DATA

The width theorem becomes a numerical exponent only after its orbit invariant is evaluated. This section evaluates it on every compact simple adjoint representation, on the defining classical representations, and on an infinite family of fundamental representations. The root and normal-form tools are classical; they are not asserted to originate in this paper. What they determine here is the dynamic width exponent under the action and calibration hypotheses of Theorem 7.3. We use the conventional root systems and highest-weight notation of [31, Sections 21–23, 27, 30–32].

9.1. All compact simple adjoint representations. Let $\mathfrak{h}$ be a compact simple real Lie algebra, with its invariant positive inner product, and let H be its connected adjoint group. Choose a maximal torus and a simple system Δ in the root system Φ of its complexification. For $I \subset \Delta$ let Φ_I be the root subsystem generated by I ; the empty subsystem has zero roots.

Theorem 9.1 (Adjoint minimum-orbit formula). *For the adjoint action on $\mathfrak{h}$,*

$$(36) \quad s(\text{Ad}) = |\Phi| - \max_{\alpha \in \Delta} |\Phi_{\Delta \setminus \{\alpha\}}|.$$

Every minimizing unit vector is conjugate to a point on a fundamental coweight ray corresponding to a maximizing deletion in (36). Conversely every such ray minimizes. The compact centralizer has Lie algebra consisting of the deleted-node Levi algebra and a one-dimensional center. The values are listed in Table 1.

Proof. Every element is conjugate into a closed Weyl chamber of the compact Cartan algebra. We identify that algebra with its real root-coordinate space. If X lies in this chamber, put $I(X) = \{\alpha \in \Delta : \alpha(X) = 0\}$. A positive root is a nonnegative integral combination of simple roots. Consequently its value on X vanishes exactly when it belongs to $\Phi_{I(X)}$. The real centralizer dimension is therefore $\text{rank } \Phi + |\Phi_{I(X)}|$, while $\dim \mathfrak{h} = \text{rank } \Phi + |\Phi|$. Hence

$$\dim(HX) = |\Phi| - |\Phi_{I(X)}|.$$

For $X \neq 0$, the simple roots span the dual Cartan space, so $I(X)$ is proper. Every proper subset is contained in a subset obtained by one deletion. If it is strictly contained, adjoining a missing simple root adds at least that root and its negative, and strictly increases the root count. Thus the largest possible centralizer occurs precisely at a maximizing single deletion. Points of the closed chamber with exactly that zero set are the positive fundamental coweight ray. Normalization chooses its unit point. This proves the formula and all the assertions about the minimizers and their centralizers.

For completeness the root counts entering the table can be obtained without an orbit-type classification. For A_ℓ , deletion of node i leaves $A_{i-1} + A_{\ell-i}$ and has $i(i-1) + (\ell-i)(\ell-i+1)$ roots. For B_ℓ, C_ℓ the analogous count is $i(i-1) + 2(\ell-i)^2$. These convex quadratic expressions are maximized at endpoints; they give the displayed counts, including the rank-two ties. For D_ℓ the chain deletions give $i(i-1) + 2(\ell-i)(\ell-i-1)$, while a spin-end deletion gives $\ell(\ell-1)$. Their maximum is $2(\ell-1)(\ell-2)$ for $\ell \geq 4$, with the familiar ties at D_4 . The exceptional diagrams have at most eight nodes: deleting them gives respectively a maximal A_1, B_3 or C_3, D_5, E_6 , and E_7 root subsystem. Their root counts are 2, 18, 40, 72, 126, whereas the original systems have 12, 48, 72, 126, 240 roots. These counts follow by reflection from their simple roots, or directly from the standard coordinate root lists. Substitution completes the table. $\square$

These are orbits of actual real vectors in the compact Lie algebra. They are not complex nilpotent orbits or projective highest-weight orbits. Confusing these objects changes the dimensions. The table is an evaluation of a classical centralizer calculation, not a claim of new classification of compact simple Lie algebras.

TABLE 1. Compact adjoint orbits and their conditional width exponents. The central one-dimensional torus in the centralizer is not a root.

Type	Largest deleted root subsystem	$s(\text{Ad})$	$s/2$
A_ℓ ($\ell \geq 1$)	$A_{\ell-1}$	2ℓ	ℓ
B_ℓ ($\ell \geq 2$)	$B_{\ell-1}$	$4\ell - 2$	$2\ell - 1$
C_ℓ ($\ell \geq 2$)	$C_{\ell-1}$	$4\ell - 2$	$2\ell - 1$
D_ℓ ($\ell \geq 4$)	$D_{\ell-1}$	$4\ell - 4$	$2\ell - 2$
G_2	A_1	10	5
F_4	B_3 or C_3	30	15
E_6	D_5	32	16
E_7	E_6	54	27
E_8	E_7	114	57

9.2. Defining and low-rank spin representations.

Proposition 9.2 (Transitive defining actions). *The defining real actions of $\text{SO}(n)$ on $\mathbb{R}^n$, $\text{SU}(n)$ on the underlying real space of $\mathbb{C}^n$, and $\text{Sp}(n)$ on $\mathbb{H}^n$ have minimum orbit dimensions*

$$n - 1, \quad 2n - 1, \quad 4n - 1,$$

respectively ($n \geq 3$, $n \geq 2$, and $n \geq 1$). Their unit-vector stabilizers are $\text{SO}(n - 1)$, $\text{SU}(n - 1)$, and $\text{Sp}(n - 1)$. The real spin representations of $\text{Spin}(3)$ and $\text{Spin}(5)$, and either complex half-spin representation of $\text{Spin}(6)$ viewed as a real representation, have minimum orbit dimensions 3, 7, 7.

Proof. Extend a unit vector to an orthonormal basis over the respective field. In the special orthogonal and special unitary cases, adjust the determinant on its orthogonal complement. This proves transitivity on the sphere. A matrix fixing the first basis vector acts freely by the indicated smaller group on its complement, giving the stabilizers and the dimensions. The standard low-rank identifications $\text{Spin}(3) = \text{Sp}(1)$, $\text{Spin}(5) = \text{Sp}(2)$ and $\text{Spin}(6) = \text{SU}(4)$ identify the stated spin modules with these defining modules (the two half-spin modules in the last case are dual). They follow also by matching the B_1 , $B_2 = C_2$, and $D_3 = A_3$ fundamental weights and dimensions [31, Sections 30–32]. Apply the preceding transitivity. $\square$

This proposition does not assert a formula for all higher spin modules. The adjoint table, including every exceptional type, and the next fundamental family have separate complete proofs.

9.3. A fundamental family whose minimum need not be a highest-weight orbit. Let $\text{SU}(n)$ act on complex skew-symmetric matrices by $A \mapsto gAg^T$. This is $\bigwedge^2 \mathbb{C}^n$, with its underlying real Euclidean structure. A fixed scalar normalization of the Hilbert–Schmidt norm does not affect orbit dimensions.

Theorem 9.3 (Second exterior powers). *For this action on its real unit sphere,*

$$(37) \quad s\left(\left(\bigwedge^2 \mathbb{C}^n\right)_{\mathbb{R}}\right) = \begin{cases} 5, & n = 3, 4, \\ 13, & n = 5, \\ 14, & n = 6, \\ 4n - 7, & n \geq 7. \end{cases}$$

For $n = 3$ and $n \geq 5$ the underlying real representation is irreducible. For $n = 4$ it is the sum of two equivalent real six-dimensional irreducibles, each with minimum orbit dimension five. At $n = 6$ a minimizing vector is a full-rank symplectic form, with stabilizer $\text{Sp}(3)$ and orbit $\text{SU}(6)/\text{Sp}(3)$; the decomposable highest-weight vector instead has orbit dimension seventeen.

Proof. We recall the skew-symmetric unitary congruence normal form and its argument. On $\mathbb{C}^n$ define the antilinear map $Fv = A\bar{v}$. Skew symmetry implies $F^2 = -AA^*$. On a positive eigenspace of AA^* with eigenvalue a^2 , F/a is antiunitary, squares to $-I$, and pairs a unit vector with an orthogonal unit vector. Inductively choosing such pairs gives blocks aJ , where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and a zero block. This is the classical unitary congruence construction [32]; the pairing argument supplies the needed special case. Although the basis change can have determinant different from one, conjugation by it preserves $SU(n)$ and therefore the dimension of the stabilizer inside $SU(n)$.

Suppose the distinct positive singular values have multiplicities $2k_1, \dots, 2k_b$, with $k = \sum_i k_i$ and $l = n - 2k$. A stabilizer preserves the eigenspaces of AA^* and commutes there with the associated quaternionic structure. Its unitary factors are therefore $Sp(k_i)$; on the kernel they form $U(l)$. The symplectic factors have determinant one, so imposing determinant one leaves $SU(l)$ on the kernel when $l > 0$. The stabilizer dimension is

$$\sum_i k_i(2k_i + 1) + \begin{cases} l^2 - 1, & l > 0, \\ 0, & l = 0. \end{cases}$$

At fixed rank, coalescing the positive singular values increases this quantity, because merging parts a, b increases it by $4ab$. Thus the smallest orbit at rank $2k < n$ has dimension

$$(38) \quad n^2 - 1 - k(2k + 1) - ((n - 2k)^2 - 1) = k(4n - 6k - 1).$$

At full rank $n = 2k$ it has dimension

$$(39) \quad n^2 - 1 - k(2k + 1) = n(n - 1)/2 - 1.$$

The quadratic in (38) is concave in k , so its minimum is at an endpoint. Comparing those endpoints with (39), when present, gives exactly (37). These minimizing strata are attained by normalized block forms, so this is not merely a dimension lower bound.

The complex module has highest weight ω_2 and dual weight ω_{n-2} ; for $n \neq 4$ these are different when $n \geq 3$. Consequently its underlying real module is irreducible of complex type. For $n = 4$ the conjugate-linear Hodge involution on two-forms has square one and commutes with $SU(4)$; its fixed real six-space and its imaginary multiple give the two equivalent real forms. The identification $SU(4) = Spin(6)$ makes each the standard real six-dimensional module, so its minimum orbit dimension is five. Alternatively the full-rank equality in (39) gives the same whole-space minimum. Finally $\dim SU(6) - \dim Sp(3) = 35 - 21 = 14$, whereas (38) at $k = 1$ gives seventeen. $\square$

Under the intrinsic action-gap and calibration hypotheses, divide the numbers in (37) by two to obtain the width exponents. In particular the $SU(6)$ fundamental module has order N^7 , not $N^{17/2}$; replacing the minimizing real-vector orbit by the highest-weight orbit would give the wrong answer. These results do not purport to classify every highest-weight representation. They compute the invariant, and its minimizing stabilizers, on the stated complete families rather than leaving those inputs as an existential rank test.

10. MATRIX AND HARMONIC REPRESENTATIONS

Theorem 4.1 identifies the correct geometric exponent without a separate estimate for every spectral type. We record its principal matrix consequences. The orbit dimensions below are classical; their role is to determine the dynamic width exponent.

Proposition 10.1 (Three conjugation families). *On the real space of traceless self-adjoint q by q matrices over $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, with inner product $\text{Re tr}(AB)$, let $\beta = \dim_{\mathbb{R}} \mathbb{F}$. The compact groups $SO(q), SU(q), Sp(q)$, respectively, act irreducibly for $q \geq 2$. The least nonzero unit-orbit dimension is $\beta(q - 1)$. Accordingly, under a full group gap, the width order in Corollary 6.3 is*

$$(40) \quad N^{(q-1)/2}, \quad N^{q-1}, \quad N^{2(q-1)},$$

respectively, in its fixed signal and fixed error ranges. The $\mathrm{SO}(2)$ case supplies no finitely supported full-gap example.

Proof. A self-adjoint matrix with distinct eigenvalue multiplicities $m_1, \dots, m_r$ has orbit dimension

$$\beta \sum_{i < j} m_i m_j = \frac{\beta}{2} \left(q^2 - \sum_i m_i^2 \right).$$

For real matrices subtract the dimensions of the orthogonal stabilizer blocks from $q(q-1)/2$; their disconnected components do not affect dimension. For complex matrices subtract $\sum m_i^2 - 1$ from $q^2 - 1$. For quaternionic matrices subtract $\sum m_i(2m_i + 1)$ from $q(2q + 1)$. These computations also identify the least orbits as real, complex and quaternionic projective spaces. A nonzero traceless matrix has at least two eigenvalues. The largest possible $\sum m_i^2$ under that restriction is $(q-1)^2 + 1$, attained by multiplicities $(1, q-1)$. This proves the minimum.

For completeness, the conjugation representation is irreducible over $\mathbb{R}$. Any nonzero invariant real subspace contains, after diagonalizing a nonzero member, a nonconstant traceless real diagonal matrix. Differences of its conjugates under coordinate interchanges give all traceless diagonal directions. Rotations in a pair of coordinate axes then give every self-adjoint off-diagonal direction; in the complex and quaternionic cases, diagonal unit phases give each real component of those entries. Signed interchanges and plane rotations can be chosen in the stated special groups. Thus the subspace is the entire traceless self-adjoint space. The general width law applies. Finally a finite measure on a torus has Fourier multipliers arbitrarily close to one by simultaneous approximation, so it has no full norm gap on its mean-zero L^2 space. $\square$

For complex matrices the projective seeds in (56) belong to a least-dimensional orbit. The signal need not be reduced to $1/(8(q^2 - 1))$ in this specialization.

Corollary 10.2 (The preserved projective width law). *Let $q \geq 2$, $0 < \rho \leq 1/10$, and $0 \leq \varepsilon < \rho/(2\sqrt{q^2 - 1})$ be fixed. For the finite projective seeds and coordinate queries of Section C, any finite command alphabet supporting a full group norm gap has*

$$W_{N,\varepsilon} = \Theta(N^{q-1}).$$

It has the fixed q -dimensional quantum realization of Proposition C.4. The explicit quantitative all-spectra estimate and the rational-projector net from the preceding revision are retained in Appendices B and C.

Proposition 10.3 (Every fixed spherical harmonic degree). *Let V_ℓ be the real irreducible $\mathrm{SO}(3)$ representation of harmonic homogeneous polynomials of degree $\ell \geq 1$, with $\dim V_\ell = 2\ell + 1$. Then $s(V_\ell) = 2$. Consequently every fixed ℓ has width $\Theta(N)$ under a full-gap command law, with the signal and query conventions of Corollary 6.3. The same six-command arithmetic alphabet in Appendix D supplies such a gap for all these representations.*

Proof. A nonzero vector has no three-dimensional stabilizer, since the representation is nontrivial irreducible. The Lie algebra $\mathfrak{so}(3)$ has no two-dimensional subalgebra: identifying its bracket with a cross product, two independent elements bracket to a third independent direction. The stabilizer dimension is therefore at most one, so every orbit has dimension at least two. A zonal spherical harmonic is fixed by the circle of rotations about its axis and has a two-dimensional orbit. This gives the minimum. The final assertion follows from Proposition E.1: the spherical norm bound controls all nontrivial $\mathrm{SO}(3)$ irreducibles, not just the defining three-dimensional action. $\square$

The degree is fixed before $N \rightarrow \infty$; no uniform-in- ℓ constants are claimed. Generic harmonic orbits can be three-dimensional, so their dimension would give the wrong exponent. In the real degree-two representation this is already the distinction between generic three-dimensional

matrix orbits and the two-dimensional projective orbit. These examples use the same general minimum-orbit theorem, not separate orbital cap computations.

11. NUMERICAL REALIZATION IS NOT CUTPOINT SIMULATION

A numerical simulator must preserve the values in (3), whereas a strict-cutpoint simulator need only preserve whether the answer probability is larger than $1/2$. The following construction makes the difference explicit in the present command model. Its stochastic embedding mechanism is classical; it is the quantitative width theorem that prevents numerical exactification at bounded width.

Proposition 11.1 (A constant-width sign simulator). *For any orthogonal action on $\mathbb{R}^D$, $D \geq 2$, there is a machine with $D + 1$ command-layer labels, common command rows, and one common query decoder for all lengths such that*

$$(41) \quad \tilde{\mathbb{P}}(B = 1 \mid x, w, j) - \frac{1}{2} = \frac{1}{2} \lambda^{N+1} \langle e_j, U_w x \rangle, \quad \lambda = 1/(2D).$$

It preserves every strict comparison of (3) with $1/2$, including equality. If $\lambda^{N+1} \leq \rho$, its worst numerical binary-TV error is at least $(\rho - \lambda^{N+1})/(2\sqrt{D})$.

Proof. Let $v_0, \dots, v_D$ be a regular simplex on the unit sphere of $\mathbb{R}^D$, centered at zero. Then

$$\sum_s v_s = 0, \quad \sum_s v_s v_s^T = \frac{D+1}{D} I, \quad b_s(z) = \frac{1 + D \langle v_s, z \rangle}{D+1}$$

are barycentric coordinates for every z in its convex hull. For $\|z\| \leq 1/(2D)$ they are all positive and sum to one. Initialize with weights $b_s(\lambda x)$. For command a use $T_a(s, r) = b_r(\lambda U_a v_s)$; these are stochastic rows, independent of N and of the epoch. Use the terminal mean $d_j(s) = \langle e_j, v_s \rangle$. The displayed barycentric identity shows inductively that the represented mean is $\lambda^{N+1} U_w x$. This proves (41) and its sign assertion. For every unit $U_w x$ some coordinate has modulus at least $1/\sqrt{D}$; binary total variation is half the difference of its means. This gives the numerical error bound. Only the label is stored. A held binary output uses a separate two-label cut. $\square$

Corollary 11.2 (Separation of numerical and sign-only width). *For every nontrivial expanding irreducible experiment in Theorem 7.3, numerical width is of order $N^{s(U)/2}$ at the stated fixed positive accuracy, but strict-cutpoint width in the same reset-command-query interface is at most $D + 1$ for all lengths simultaneously. For complex conjugation the numerical order is N^{q-1} , while the sign-only command width is at most q^2 .*

Proof. Combine Proposition 11.1 with the corresponding numerical width theorem. Here $D = q^2 - 1$ in the last assertion. The number counts command-layer labels, not the total state count of a language recognizer additionally parsing reset and query syntax. $\square$

11.1. The current quantum-automata comparison. We fix the version before comparing its theorem. The primary arXiv submission record inspected on September 26, 2026 lists version 1 dated April 8 and version 2 dated August 27, 2026. In version 1 [24], Proposition 3.1 is the q^2 -dimensional mixed-state linearization, Theorem 3.2 gives $2k + 6$ PFA states from a k -state GFA, and Corollary 3.3 gives $2q^2 + 6$. In the explicitly versioned version 2, the title and theorem numbering change. The PDF's printed pages 4 and 5 contain, respectively, Proposition 3.1 and Theorem 3.2 with the improved constants used below. These are distinct primary versions, not interchangeable theorem numbers. We make no inference about why an earlier access exposed only one version.

Chen–Wu’s version-2 generalized-automaton conversion [23, Proposition 3.1] maps a k -dimensional real zero-cutpoint generalized finite automaton to a probabilistic automaton with $k + 1$ states. For its parameters $c, \delta > 0$ the conversion has

$$(42) \quad f_{\text{PFA}}(w) - 1/(k + 1) = \delta c^{|w|} f_{\text{GFA}}(w).$$

Their Theorem 3.2 consequently gives $q^2 + 1$ states for strict-cutpoint simulation of a q -state general one-way QFA. The second version of that paper improves the larger bound in its original version. Mixed-state linearization is exact before the positive embedding; the factor in (42) is the distinction relevant here. The later paper [25, Proposition 4 and Theorems 11–12] uses the related conversion to compare hybrid quantum and classical language state costs. These are results about cutpoint languages, not fixed nonvanishing numerical accuracy.

For clarity, the algebra behind (42) is that a uniform-row matrix Q annihilates a zero-sum coordinate subspace, while a scaled matrix cK_a carries the desired signed dynamics there. Matrices $Q + cK_a$ are made stochastic by choosing c small. After N commands the signed deviation is multiplied by c^N . A length-dependent decoder attempting to divide by this factor need not lie in $[-1, 1]$ and is not a stochastic readout. More generally, a stochastic post-processing cannot increase total variation. Allowing redesign at every horizon does not overcome the numerical lower bound, which already permits that redesign.

A prefix–suffix sign or rank argument alone cannot substitute for the occupation proof. The matrix of centered target responses has entries $\langle \rho U_p x, U_s^* e_j \rangle$ and ordinary rank at most D at every split, independently of the horizon. Adding the constant probability column gives rank at most $D + 1$. The prepare–test sign-rank and shattering statements in [25] control a threshold relation in this fixed operator space; they do not force the numerical hidden-width growth proved here. Conversely the present argument gives no growing lower bound for the sign-only task, as Proposition 11.1 demonstrates.

Chen’s subsequent work [26], including its invariant-algebra simulation results and Section 7’s stabilizer-orbit capacity analysis, also uses strict-cutpoint behavioral equivalence. Its dimensions and symmetry constraints belong in this comparison. Orbit dimension as $\dim G - \dim G_v$, and quantum operator-space linearization, are not claimed new here. Our invariant first takes the *minimum over all nonzero directions in each irreducible representation*, then the *maximum across active constituents*; it governs a growing fixed-accuracy numerical width as the word length grows. These are different optimization and asymptotic variables. No assertion of exhaustive priority clearance is made from this comparison.

12. GEOMETRIC, PROBABILISTIC AND COMPUTATIONAL BOUNDARIES

The new theorem concerns a growing execution constrained by a single retained label. It neither changes the classical dimension of an orbit nor identifies numerical simulation with a threshold language.

12.1. What the covering exponent does and does not prove. For a smooth compact s -manifold, covering by $O(h^{-s})$ small balls is classical. Iacobelli’s manifold quantization theorem [27, Theorem 1.4] concerns the minimum integrated r th-power distance from a *fixed measure* to a codebook of k points, with its $k^{-r/s}$ asymptotics under the stated hypotheses. The codebook in that problem sees one sample from that measure; there are no stochastic command rows to be implemented at all layers. Theorem 6.2 uses the classical covering count, a quadratic support loss, and an explicit stochastic realization. Its new lower partner is not a static quantization lower bound. The conditional means of a hidden machine generally leave the seed orbit, and neither their distribution nor their orbit type is fixed. Theorem 4.1 controls all those directions at once; the entropy budget then charges every small-state epoch to one finite quadratic loss. This is the mechanism that prevents hidden lifts from outperforming the geometric synthesis order.

Orbit convex hulls and their representation-theoretic geometry are classical objects [21, 28]. The covariance identity and local covering calculation are reproved here to fix constants and hypotheses, not claimed as new geometric discoveries. Our minimum-orbit theorem does not give local volume asymptotics uniformly for each spectrum; only its uniform exponent is sharp. The elementary finite-rank chart argument also explains why no separate spectral projector calculation is needed for a new representation. The earlier projector argument is retained because it gives an explicit, albeit large, constant in its matrix family.

12.2. Positive rank and quantum dimension. At a fixed cut, a numerical classical realization factors the matrix of future response probabilities through K_t positive normalized state continuations. With one output alphabet this is a nonnegative factorization; with several selectable queries, each decoder block must normalize. Quantum prepare–measure probabilities instead admit positive-semidefinite factorizations. These static dimension relations are classical [29]. They do not by themselves supply common transition rows connecting independently chosen factorizations at successive cuts. In the projective experiment a fixed q -dimensional quantum register supplies all layers, while the compatible classical numerical width grows with N . No new quantum operation, seed reset, or quantum-memory model is introduced by that comparison.

Proposition 12.1 (Probabilistic ordered programs). *Definition 2.1 is equivalently a nonuniform ordered read-once layered program with stochastic command-labelled edges and real terminal columns. A selected column can be replaced by a Bernoulli acceptance row without changing any command-layer width; a retained final answer needs at most two labels.*

Proof. Identify $T_{t,a}(s, s')$ with the outgoing edge weight at layer t . Summing path probabilities gives (4), and conversely those edge weights define the stochastic rows. At terminal label s the selected query accepts with probability $(1 + d_j(s))/2$. $\square$

The Fourier-growth and pseudorandomness results in [15, 16, 9] have their stated deterministic, regular or permutation hypotheses and different objectives. The dominant action used by the converse has a norm gap; hidden reset rows and nonregular matrices remain allowed. Width is not total branching-program size, and real edge tables are not a uniform finite-bit implementation.

12.3. Expansion and implementation. Expansion in the present main theorem is a property of the actual command action on an activated irreducible space. It is unchanged by adding unused presenting factors. Theorem 7.3 requires it only on one dominant active type; Proposition 8.1 even allows a faithfully acting circle factor with no regular gap. Full effective-group expansion remains a sufficient certificate via Proposition 3.1. The algebraic alphabets in Proposition C.3 and in the mixed example use the deep external theorems [2, 3]; their constants are qualitative. Density or a finite harmonic calculation is not itself such a certificate. The optional numerical six-gate result uses exactly the separate external inequality (LPS5). Its original-source limitation is recorded in Appendix E.

Only one query is answered. A request for jointly independent answers to all queries changes the channel and need not satisfy the maximum law of Theorem 7.3. The model charges branch identifiers, but not the description length or construction of the row tables. Rational net coordinates do not make all transition weights rational when commands are merely algebraic. Carathéodory sparsity bounds successors, not table size. The separate fair-bit compilers in the preserved development incur further states and possibly error; they do not equate atomic width with bit complexity.

12.4. Conclusions. The effective action and the query-calibrated isotypic activation remove presentation and multiplicity choices from the width theorem. Arbitrary finite seeds first determine their invariant span. A single expanding type of maximal minimum-orbit dimension then determines

the matched exponent; other genuinely acting types may fail to expand. The upper realization charges its branch labels, while the lower bound continues to use the actual entire hidden register.

The root-deletion and singular-value calculations identify minimum orbits on the stated complete representation families. They do not classify all highest weights. Likewise the width law does not specify finite integer optima, sharp leading constants, arbitrary vanishing signals, nongapped dominant types, or uniform computational resources. Numerical probability preservation and strict-cutpoint recognition are separated by an explicit constant-width attenuating machine. The version-pinned comparison makes this semantic boundary independent of which Chen–Wu upper constant is quoted.

APPENDIX A. A MAXIMUM LAW FOR ORTHOGONAL SUMS

Let

$$(43) \quad V = V_0 \oplus V_1 \oplus \cdots \oplus V_m$$

be an orthogonal decomposition, where $V_0 = V^G$ and each V_i , $i \geq 1$, is a nontrivial real irreducible representation. Multiplicities are allowed. Put $d_i = \dim V_i$, $D = \dim V$ and

$$(44) \quad \sigma(U) = \max_{1 \leq i \leq m} s(U|_{V_i}), \quad \sigma(U) = 0 \quad \text{if } m = 0.$$

Equivalent irreducible orthogonal representations have isomorphic orbits. Thus $\sigma(U)$ does not depend on the chosen decomposition of an isotypic summand. In general it is *not* the smallest orbit dimension on the sphere of the whole reducible representation.

For the lower bound assume that the finite unit seeds span V and write

$$(45) \quad a_i = \max_{x \in \mathcal{X}} \|\pi_i x\| > 0, \quad a_* = \min_{1 \leq i \leq m} a_i,$$

where π_i is orthogonal projection; set $a_* = 1$ if $m = 0$. In a basis-adapted seed set $\{\pm e_j\}$ all a_i equal one. The query basis need not respect the decomposition.

Theorem A.1 (Irreducible maximum principle). *Suppose a law on the finite command alphabet has a full group norm gap. Let the seeds span V , let $0 < \rho \leq \min\{1/10, 1/(8D)\}$, and, when $m > 0$, let $0 \leq \varepsilon < \rho a_*/(2\sqrt{D})$. Then*

$$(46) \quad W_{N,\varepsilon} = \Theta(N^{\sigma(U)/2}).$$

The constants may depend on the representation, command law, seed set, signal and error, but not on N . If $m = 0$, an exact constant-width machine exists. The upper realization uses common command rows and only one selected terminal numerical query.

Proof. First fix $i \geq 1$ and a seed x with $\|\pi_i x\| = a_i$. Track the unit-vector process $Y_t^{(i)} = U_{a_t} \cdots U_{a_1} \pi_i x / a_i$ in V_i and its conditional mean given the *actual whole-machine state*. Form the ambient vector $d = (d_1, \dots, d_D)$ from the terminal columns in the proof. It has norm at most $\sqrt{D}$. The prescribed row means and the uniform error give

$$\mathbb{E}\langle Y_N^{(i)}, \pi_i d(S_N) \rangle \geq \rho a_i - 2\sqrt{D}\varepsilon.$$

Conditioning on S_N bounds the left side above by $\sqrt{D}\mathbb{E}\|\mathbb{E}[Y_N^{(i)} | S_N]\|$. Hence the terminal conditional amplitude is at least $\rho a_i / \sqrt{D} - 2\varepsilon > 0$. The inherited entropy occupation argument applies in the representation V_i to these conditional means, even though the hidden label also serves every other constituent. Theorem 4.1 supplies its exponent $s(U|_{V_i})$. It follows that $W_{N,\varepsilon} \geq c_i N^{s(U|_{V_i})/2}$. Choose a constituent attaining (44).

For the upper bound include only nonzero summands and set $w_i = d_i/D$. Choose a branch i at initialization with probability w_i , independently of the seed. The branch identifier is part

of the state label, not a free future input. In branch $i \geq 1$ use the minimal-orbit polytope of Theorem 6.2. Its initial represented mean is prescribed to be

$$(\rho/w_i)\pi_i x.$$

Its norm is at most $\rho D/d_i \leq 1/(8d_i)$, so the initialization proof there applies also to this possibly subunit vector. Subsequent rows are the common command rows from that construction. At a query j output the mean $\langle e_j, a_N v_{i,s} \rangle$. The conditional represented vector in the branch is $(\rho/w_i)U_w \pi_i x$; multiplying by w_i and summing gives exactly $\rho U_w x$. All terminal means are bounded by $2/7$.

In the invariant branch, if present, use the fixed vectors $\{\pm e : e \text{ in an orthonormal basis of } V_0\}$ scaled by $1/4$, with identity transitions. Their convex hull contains a ball of radius $1/(4\sqrt{d_0})$, which contains the required initial vector of norm at most $1/(8d_0)$. Its contribution is exact with $2d_0$ labels. The entire register is the disjoint union of these branch labels. Its cardinality is at most

$$2d_0 + \sum_{i=1}^m C_i N^{s(U|V_i)/2} = O(N^{\sigma(U)/2}).$$

No product of independently correct outputs is required. If $m = 0$ the invariant construction proves the asserted constant bound. The fixed two-label answer cut does not change the orders. $\square$

The original full-group maximum principle is retained here in its original sufficient range. Theorem 7.3 strengthens it by using the active span, canonical activation and only one dominant action gap; its proof of the current main theorem appears in Section 7.

The maximum in (44), rather than a sum, is substantive. For the direct product action of $\text{SO}(3) \times \text{SU}(3)$ on $\mathbb{R}^3 \oplus \text{Herm}_0(3)$, the two least orbit dimensions are two and four. With a full-gap product command law and the above signal conventions the label order is N^2 , not N^3 ; a generic product orbit and a joint-output product task describe different quantities. Adding a fixed subspace creates zero-dimensional whole-space orbits but does not remove this N^2 requirement. The proof explains both phenomena by paid branch labels and a constituent-wise converse.

Proposition A.2 (All separate cuts have constant positive minimum). *For any orthogonal command family on $\mathbb{R}^D$, seeds $\mathcal{X} = \{\pm e_1, \dots, \pm e_D\}$, and $0 < \rho \leq 1/D$, the normalized residual row rank and the separate positive minimum are both $D + 1$ at every internal cut. The separate minimum permits unrestricted memory at the other cuts. The assertion requires no group gap and does not assert simultaneous attainment.*

Proof. For a vector z , its remaining response on suffix v and query j has mean $\langle e_j, U_v z \rangle$. The resulting affine map from z to the normalized response array is injective: fix one suffix and vary the coordinate query; U_v is invertible. At a fixed cut, fix one preceding word w and vary the seeds. The vectors $\{\pm \rho U_w e_j\}$ affinely span $\mathbb{R}^D$. Thus the normalized residual arrays have row rank $D + 1$, giving the positive lower bound.

A regular unit simplex in $\mathbb{R}^D$ contains the ball of radius $1/D$. Every actual represented vector at that cut has norm ρ and is therefore a convex combination of its $D + 1$ vertices. A vertex v_s defines a legal normalized continuation, since $|\langle e_j, U_v v_s \rangle| \leq 1$ for every suffix and query. These are $D + 1$ positive generators of all cut rows. To embed the factorization in a whole machine, retain the complete finite prefix until the selected cut, sample its convex coordinates there, then retain the chosen label together with the remaining command prefix. There is no cap at those other cuts. The resulting finite machine attains $D + 1$ at the selected cut. This verifies the operational, not merely amputated-matrix, meaning of the separate minimum. $\square$

APPENDIX B. UNIFORM GEOMETRY OF HERMITIAN CONJUGATION

Fix $q \geq 2$ and let $V = \text{Herm}_0(q)$ be the real space of traceless Hermitian q by q matrices, with inner product $\text{tr}(AB)$. Its dimension is $D = q^2 - 1$. Unitary conjugation is orthogonal on this space. The pure-state orbit has dimension $2(q-1)$, but normalized hidden centroids can have arbitrary spectra. We therefore prove a bound on *every* unit sphere orbit.

B.1. A Grassmann small-ball estimate. Let P_0 be a rank- k orthogonal projector in $\mathbb{C}^q$, $1 \leq k < q$, and let P be a Haar-distributed rank- k projector. Put $m = k(q-k)$ and

$$(47) \quad b_{q,k} = \frac{1}{m!} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)}.$$

Lemma B.1 (Projector balls). *For $0 < R \leq 1$,*

$$(48) \quad \mathbb{P}(\|P - P_0\|_{\text{HS}} \leq R) \leq b_{q,k} R^{2k(q-k)}.$$

Proof. By invariance take P_0 onto the first k coordinates. Apart from a null set, a Haar subspace is the graph of $Z \in \mathbb{C}^{(q-k) \times k}$. Its density with respect to complex Euclidean Lebesgue measure is

$$(49) \quad \pi^{-m} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)} \det(I + Z^*Z)^{-q}.$$

For completeness, take a q by k matrix of independent standard complex Gaussian entries and split it into a square top block A and bottom block B . Its column space is Haar and $Z = BA^{-1}$ almost surely. The change $B = ZA$ has real Jacobian $|\det A|^{2(q-k)}$. Integrating A with weight $\exp[-\text{tr} A^*(I + Z^*Z)A]$ gives the determinant factor $\det(I + Z^*Z)^{-q}$. Complex Gaussian Gram-Schmidt gives $\mathbb{E}|\det A|^{2(q-k)} = \prod_{j=1}^k \Gamma(q-k+j)/\Gamma(j)$, since its squared orthogonal lengths have gamma shapes $1, \dots, k$. The powers of π give (49).

If θ_i are the principal angles, then $\|P - P_0\|_{\text{HS}}^2 = 2 \sum_i \sin^2 \theta_i$ and $\|Z\|_{\text{HS}}^2 = \sum_i \tan^2 \theta_i$. On the event in (48) each $\sin^2 \theta_i \leq R^2/2$, whence

$$\|Z\|_{\text{HS}}^2 \leq \frac{R^2}{2 - R^2} \leq R^2.$$

The density (49) is bounded by its value at zero. Multiply this value by the volume $\pi^m R^{2m}/m!$ of a real $2m$ -dimensional ball. This proves the assertion. $\square$

Define the explicit finite constant

$$(50) \quad A_q = \max_{1 \leq k < q} \left\{ \max(1, b_{q,k}) [9q(q-1)^2]^{k(q-k)} \right\}.$$

It is rational and at least one; no optimality is claimed.

Theorem B.2 (All-spectra orbital anti-concentration). *For all $A, B \in \text{Herm}_0(q)$ of Hilbert-Schmidt norm one and all $0 < r \leq 1$,*

$$(51) \quad m_{\text{SU}(q)}\{U : \|UAU^* - B\|_{\text{HS}} \leq r\} \leq A_q r^{2(q-1)}.$$

In particular, the estimate is uniform at repeated eigenvalues and over mixed-state centroid directions. No lower bound on each eigenvalue gap is assumed.

Proof. Let $\lambda_1 \geq \dots \geq \lambda_q$ be the eigenvalues of A . Trace zero and $\sum_i \lambda_i^2 = 1$ imply $\lambda_1 - \lambda_q \geq q^{-1/2}$. Hence some k has

$$(52) \quad \lambda_k - \lambda_{k+1} \geq \delta_q := \frac{1}{\sqrt{q}(q-1)}.$$

For two conjugates C, D of A , let P_C, P_D be their top- k spectral projectors. Eigenvectors u_i of C and v_j of D obey

$$(\lambda_i - \lambda_j)\langle u_i, v_j \rangle = \langle u_i, (C - D)v_j \rangle.$$

Summing squares for $i \leq k < j$ and using equal projector ranks gives

$$(53) \quad \|P_C - P_D\|_{\text{HS}}^2 = 2 \sum_{i \leq k < j} |\langle u_i, v_j \rangle|^2 \leq 2\delta_q^{-2} \|C - D\|_{\text{HS}}^2.$$

If the event in (51) is empty there is nothing to prove. Otherwise choose one conjugate D in that event. Every other C in the event has $\|C - D\|_{\text{HS}} \leq 2r$. The image P_C is a Haar rank- k projector and lies within $2\sqrt{2}\delta_q^{-1}r$ of P_D . Set $c_q = 3\sqrt{q}(q-1)$, which is larger than this coefficient. If $c_q r \leq 1$, Lemma B.1 gives a bound $b_{q,k}(c_q r)^{2k(q-k)}$. Since $k(q-k) \geq q-1$ and $r \leq 1$, this is at most the right side of (51). If $c_q r > 1$, then $A_q r^{2(q-1)} \geq c_q^{2(q-1)} r^{2(q-1)} > 1$, so the trivial probability bound suffices. Haar conjugation under $\text{SU}(q)$ and under $U(q)$ has the same law, completing the proof. $\square$

The proof isolates one uniformly separated spectral cut rather than requiring a simple spectrum. This is why pure-state and highly mixed centroids are handled by the same constant. The smallest possible nontrivial Grassmann exponent is $2(q-1)$.

Corollary B.3 (Conjugation occupation). *For conjugation commands satisfying a group gap g_t , any hidden register with $q_N \geq \kappa > 0$ obeys*

$$(54) \quad \sum_{K_t \leq k} g_t \leq C_q \kappa^{-3-1/(q-1)} k^{1/(q-1)}, \quad C_q = 2048(q^2 - 1)(16A_q)^{1/(q-1)}.$$

For a constant gap $g > 0$ and wordwise error ε with $\kappa = \rho/\sqrt{q^2 - 1} - 2\varepsilon > 0$,

$$(55) \quad W_{N,\varepsilon} \geq \left(\frac{g\kappa^{3+1/(q-1)}}{C_q} N \right)^{q-1}.$$

Proof. Apply Lemma 5.1, Theorem B.2 and Theorem 5.5, then use Lemma 2.2. $\square$

A group gap is stronger than a gap only on functions on the pure-state homogeneous quotient. The proof genuinely needs uniform control of the mixed-centroid directions, and does not silently replace this hypothesis by the weaker quotient gap.

APPENDIX C. SHARP MEMORY ON COMPLEX PROJECTIVE ORBITS

Write $r_q = \sqrt{1 - 1/q}$ and associate to a rank-one orthogonal projector P the unit vector

$$(56) \quad Y(P) = \frac{P - I/q}{r_q} \in \text{Herm}_0(q), \quad \mathcal{K}_q = \text{conv}\{Y(P) : \text{rank } P = 1\}.$$

The body $\mathcal{K}_q$ consists of normalized centered density matrices. It is invariant under conjugation and contains zero. Its extreme orbit is $\mathbb{CP}^{q-1}$. Fix an orthonormal real basis $F_1, \dots, F_D$ of $\text{Herm}_0(q)$ and use its coordinate queries in (3). One may take as seeds the q^2 rank-one projectors onto e_i , $(e_i + e_j)/\sqrt{2}$ and $(e_i + ie_j)/\sqrt{2}$, $i < j$. Their centered matrices affinely span $\text{Herm}_0(q)$. All seed and query alphabets are fixed when q is fixed.

C.1. A projective enclosure with quadratic loss. For rank-one projectors use the distance $d(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, the sine of their principal angle.

Lemma C.1 (Finite projective synthesis). *For $0 < h \leq 1$, there is an h -net $\mathcal{V}_h$ of rank-one projectors with at most*

$$(57) \quad q(2\lceil 2\sqrt{2(q-1)/h} \rceil + 1)^{2(q-1)}$$

elements, and the entries of all these projectors can be chosen in $\mathbb{Q} + i\mathbb{Q}$. If $qh^2 < 1$, then

$$(58) \quad (1 - qh^2)\mathcal{K}_q \subseteq \text{conv}\{Y(P) : P \in \mathcal{V}_h\} \subseteq \mathcal{K}_q.$$

Proof. Every complex line has a representative with a largest coordinate scaled to one and the real and imaginary parts of all other coordinates in $[-1, 1]$. In each of the q charts round these $2(q-1)$ real coordinates to the grid of step $1/M$, where $M = \lceil 2\sqrt{2(q-1)/h} \rceil$. If v, w are the original and rounded representatives, their norms are at least one and $\|v - w\| \leq \sqrt{2(q-1)}/(2M)$. Normalization changes their distance by at most a factor two, and the sine distance is at most the distance of these normalized representatives. Thus the stated set is an h -net. The projector $ww^*/\|w\|^2$ has rational real and imaginary entries, and the grid count is (57).

For a nonzero traceless Hermitian H , let λ_1 and λ_q be its largest and smallest eigenvalues. A net projector within h of a top eigenline satisfies

$$\text{tr}(HP) \geq \lambda_1 - (\lambda_1 - \lambda_q)h^2 \geq (1 - qh^2)\lambda_1.$$

Indeed $-\lambda_q \leq (q-1)\lambda_1$ by trace zero. The support function of $\mathcal{K}_q$ at H is λ_1/r_q . The support-function criterion for convex containment proves (58). It applies to every direction, not just a finite set of tested queries. $\square$

Theorem C.2 (A finite-alphabet exact upper machine). *For every finite unitary command alphabet and $0 < \rho \leq 1/10$, the experiment on (56) has an exact machine of width $O_q(N^{q-1})$. For each N , all command transition matrices can be chosen independent of the epoch. Each row uses at most q^2 successors. The final decoder may depend on N .*

Proof. Take $h = (8qN)^{-1/2}$ and $\eta = 1 - 1/(8N)$ in Lemma C.1. Let V be the resulting finite set of centered vectors, and put $a_t = \eta^{-t}/4$. A label i represents $a_t Y(P_i)$ at epoch t . This is a proof description of a label; no matrix is stored in the register.

Since $\rho/a_0 = 4\rho \leq 2/5 < \eta$ and $0 \in \mathcal{K}_q$, any seed $\rho Y(P)$ is a convex combination of $a_0 V$. For each label i and each command U , the vector $\eta UY(P_i)U^*$ lies in $\text{conv } V$. Choose a convex representation and use its weights as the stochastic transition row. The choices depend on the command and label but not on t . Multiplying by a_{t+1} shows that the expected represented vector is updated by exact conjugation, since $a_{t+1}\eta = a_t$.

Induction gives expected vector $\rho Y(U_w P U_w^*)$. The terminal column for query j is $a_N \text{tr}(F_j Y(P_i))$. It is legal because $|\text{tr}(F_j Y(P_i))| \leq 1$ and Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7$. Sampling a binary answer with this mean proves every conditional probability in (3) exactly. Carathéodory's theorem in real dimension $D = q^2 - 1$ permits q^2 successors. The number in (57) is $O_q(h^{-2(q-1)}) = O_q(N^{q-1})$. $\square$

The upper argument uses a quadratic support loss along the projective orbit. Applying a net on the full ambient sphere would instead give the larger exponent $(q^2 - 2)/2$.

C.2. Fixed finite algebraic alphabets. For $1 \leq j < q$ let $R_j, D_j \in \text{SU}(q)$ act as the identity outside the adjacent coordinate pair $(j, j+1)$ and on that pair as

$$(59) \quad R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} \zeta & 0 \\ 0 & \bar{\zeta} \end{pmatrix}, \quad \zeta = \frac{3 + 4i}{5}.$$

The alphabet has $4(q-1) + 1$ symbols: $\{I, R_j^{\pm 1}, D_j^{\pm 1} : 1 \leq j < q\}$. Assign probability $1/2$ to I and $1/[8(q-1)]$ to each other symbol.

Proposition C.3 (Qualitative group expansion). *This command law has a norm gap on $L_0^2(\text{SU}(q))$ for every fixed q . Its entries and probabilities are algebraic. No numerical gap value is asserted for this family.*

Proof. The number ζ is not a root of unity: $\zeta + \zeta^{-1} = 6/5$ is rational but not an algebraic integer. The powers of R_j and D_j therefore have closures containing two distinct one-parameter subgroups in the corresponding embedded $\text{SU}(2)$. Their Lie brackets generate that $\mathfrak{su}(2)$. The adjacent embedded Lie algebras generate $\mathfrak{su}(q)$: adjacent elementary off-diagonal directions produce all other off-diagonal directions by brackets, and their self-brackets produce the traceless diagonal directions. The closed generated subgroup is consequently $\text{SU}(q)$.

The adjoint entries are algebraic. The spectral-gap theorem of Benoist–de Saxcé [2, Theorem 1.2] applies to an adapted algebraic measure on this compact connected group with simple Lie algebra. Equivalently, one may use the algebraic unitary result of Bourgain–Gamburd [3]. The symmetric average has an upper gap; weight $1/2$ at the identity makes its spectrum nonnegative. Its mean-zero operator norm is therefore strictly below one. This is the external full-group theorem, not a calculation in the defining q -dimensional representation. $\square$

Proof of Corollary 10.2. Corollary B.3 gives the lower order, for every fixed positive calibration κ . Theorem C.2 gives the matching exact upper order. Proposition C.3 supplies fixed finite alphabets. The quantum assertion follows from the next proposition. Taking binary logarithms gives the label-bit formula. $\square$

Proposition C.4 (A fixed quantum realization). *The projective experiment has an exact quantum realization on $\mathbb{C}^q$ using fixed seed resets, the prescribed unitary commands and fixed query measurements. It is independent of N .*

Proof. On seed P reset the density matrix to

$$\sigma_P = I/q + \rho Y(P) = (1 - \rho/r_q)I/q + (\rho/r_q)P.$$

This is positive and has trace one, since $0 < \rho \leq 1/10 < r_q$. Commands act by conjugation. For query j , use effects $(I + F_j)/2, (I - F_j)/2$. Hilbert–Schmidt normalization implies $\|F_j\|_{\text{op}} \leq 1$, so these are positive effects summing to I . Their outcome mean is $\text{tr}(F_j \sigma_{U_w P U_w^*}) = \rho \text{tr}(F_j Y(U_w P U_w^*))$. This is precisely (3). The seed reset is a channel, not a unitary; only one query is measured. $\square$

The comparison is a fixed q -dimensional quantum register versus $(q-1) \log_2 N + O(1)$ classical label bits. Dimension or entropy advantages for quantum models are prior phenomena [6, 17, 5]. The assertion here is the proved classical width order for this complete worst-word numerical behavior. It does not request jointly independent answers to all queries or optimize arbitrary adaptive experiments.

APPENDIX D. AN EFFECTIVE SIX-COMMAND RATIONAL EXAMPLE

The preceding group-gap application is qualitative. We now choose a separate arithmetic alphabet for which a classical theorem supplies a numerical infinite-dimensional norm bound. This does not evaluate the gap of the earlier five-command example.

Let R_x, R_y, R_z be rotations about the coordinate axes with cosine $-3/5$ and sine $4/5$; explicitly,

$$(60) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & 4/5 & -3/5 \end{pmatrix}, \quad R_y = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & -3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the fixed six-letter alphabet $\mathcal{A}_V = \{R_x^{\pm 1}, R_y^{\pm 1}, R_z^{\pm 1}\}$ and its uniform law. These are the adjoint rotations of the classical norm-five quaternions $1 \pm 2\mathbf{i}$, $1 \pm 2\mathbf{j}$, $1 \pm 2\mathbf{k}$, normalized by $\sqrt{5}$.

Proposition D.1 (External arithmetic gap certificate). *For $P_V f = (1/6) \sum_{R \in \mathcal{A}_V} f(R^{-1} \cdot)$ on $L_0^2(\mathbb{S}^2)$,*

$$(61) \quad \|P_V\| \leq \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}.$$

Consequently $1 - \|P_V\|^2 \geq 4/9$.

Proof. This is the $p = 5$ specialization of the Lubotzky–Phillips–Sarnak spherical Hecke bound [10, 11]. In its unnormalized convention the $p + 1$ norm- p quaternion representatives act by a sum of rotations whose mean-zero norm is at most $2\sqrt{p}$. The representatives have positive odd real coordinate and even other coordinates when $p \equiv 1 \pmod{4}$; at $p = 5$ they are exactly the six displayed ones. Appendix E records the operator normalization and its source boundary. The coauthor account [12, Section 3.2, the V -gate example] spells out these representatives and their Hecke interpretation. The unnormalized spherical norm convention is also stated in [13, Theorem 3(ii)]. Dividing by six gives (61). Passing from quaternions to adjoint rotations identifies $\mathbb{S}^2$ as the homogeneous quotient; inverse signs do not affect this symmetric average. The arithmetic bound is invoked as an external theorem, not reproved by finite harmonic tests. $\square$

Use six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, amplitude $\rho = 1/10$ and three final coordinate queries in (3).

Theorem D.2 (Numerical finite-horizon width certificate). *For $0 \leq \varepsilon < 1/(20\sqrt{3})$ and every integer $N \geq 1$,*

$$(62) \quad \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N \leq W_{N,\varepsilon}(\mathcal{A}_V) \leq W_{N,0}(\mathcal{A}_V) < 150N.$$

The upper machine has exact rational command and decoder rows, common command matrices, and an N -dependent decoder. At zero error every internal selected cut has separate positive minimum four, while

$$(63) \quad W_{N,0} \geq \frac{N}{19\,906\,560\,000}.$$

The constants are proof constants, not sharp asymptotic constants.

Proof. For the standard $\text{SO}(3)$ action the orbit of every unit vector is uniform on $\mathbb{S}^2$. A chordal cap of radius $r \leq 1$ has probability $r^2/4 \leq r^2$, so take $D = 3, s = 2, A = 1$ in Theorem 5.5. The sphere gap transfers to $L^2(\mathbb{R}^3)$ by polar coordinates. Thus $B_{3,1,2} = 98304$, and the squared gap $4/9$ gives the lower bound in (62). At $\varepsilon = 0$ the fourth power of $1/(10\sqrt{3})$ is $1/90000$, which yields (63).

Here is an explicit cardinality estimate for the upper construction. Take both rational stereographic charts

$$t \mapsto \frac{(2t_1, 2t_2, \pm(1 - \|t\|^2))}{1 + \|t\|^2}, \quad t \in [-1, 1]^2,$$

with a grid of step $1/M$ and $M = \lceil 2\sqrt{2N} \rceil$. Every hemisphere is covered by its chart on the unit disk. Each chart is 2-Lipschitz, so its grid images form a sphere net of chordal radius at most $\sqrt{2}/M \leq 1/(2\sqrt{N})$. Its convex hull contains η times the unit ball, with $\eta = 1 - 1/(8N)$, by the support function identity $u \cdot v = 1 - \|u - v\|^2/2$. There are at most $2(2M + 1)^2 \leq (82 + 48\sqrt{2})N < 150N$ labels. Use scales $a_t = \eta^{-t}/4$, initialization representing ρx , rows representing $\eta R u$ in this hull, and decoder $a_N e_j \cdot u$. The verification is the induction in Theorem C.2. All data are rational; each nonempty rational convex system has rational weights, with at most four successors. The final decoder is bounded by $2/7$.

At a selected internal cut, fix any common command prefix. The six seed images affinely span $\mathbb{R}^3$, so the normalized continuation rank is four; future coordinate queries still separate these dimensions after any common suffix. For an upper factorization take a regular tetrahedron containing the radius- ρ ball and having circumradius $3\rho < 1$. Its four vectors give legal continuations under all future rotations. One embeds this selected-cut factorization in a complete machine by allowing uncapped raw histories elsewhere. Therefore the separate minimum is exactly four. This does not give a four-state machine at all cuts simultaneously. $\square$

For example, (63) exceeds four once $N \geq 79\,626\,240\,001$. The size of this number illustrates the conservative constants; the benefit is that no unspecified spectral constant is needed for this alphabet.

The associated qubit operations are $(I - 2i\sigma_j)/\sqrt{5}$ and their inverses, with seed Bloch vector $x/10$ and final Pauli query. They realize the numerical experiment exactly, independently of N . This quantum construction and the arithmetic gates are classical; (62) is the finite-register consequence. The earlier five-gate alphabet remains qualitatively gapped with a non-evaluated constant, and is retained unchanged in the supporting material.

APPENDIX E. THE ARITHMETIC OPERATOR AND ITS NORMALIZATION

This appendix fixes the exact external input used in Proposition D.1, then proves the remaining representation and normalization steps. Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ for quaternion units and

$$\mathcal{Q}_5 = \{1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}\}, \quad r_q(z) = qzq^{-1} \quad (z \in \text{Im } \mathbb{H}).$$

The required external LPS spherical Hecke inequality, specialized to this representative convention, is

$$\text{(LPS5)} \quad \left\| \sum_{q \in \mathcal{Q}_5} f(r_q^{-1} \cdot) \right\|_{L^2(\mathbb{S}^2)} \leq 2\sqrt{5} \|f\|_{L^2(\mathbb{S}^2)}, \quad \int f = 0.$$

This is an imported arithmetic theorem, not a finite-dimensional calculation. The original sources are [10, 11]; the unnormalized norm theorem is explicitly restated in [13, Theorem 3(ii), p. 8 of the preprint], while the quaternionic and Hecke representative conventions are discussed in [12, Section 3.2]. The restated existence theorem alone would not identify an arbitrary six rotations: the representative convention in (LPS5) is essential.

Proposition E.1 (Parity, quotient and full spherical spectrum). *With the external input (LPS5), the uniform average over the six rotations in (60) has mean-zero norm at most $\sqrt{5}/3$ both on $L^2(\mathbb{S}^2)$ and on $L^2(\text{SO}(3))$. Its squared gap is at least $4/9$. The conclusion does not assert the same numerical gap for all of $L_0^2(\text{SU}(2))$.*

Proof. An integral quaternion of norm five has one coordinate of magnitude one, one of magnitude two, and two zero coordinates. There are $4 \cdot 3 \cdot 4 = 48$ such quaternions. Multiplication by a Lipschitz unit in $\{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ places the odd coordinate in the real position and makes it positive. This gives a unique representative with positive odd real part and even other coordinates, namely one of the six members of $\mathcal{Q}_5$. This specifies the parity class; it is not an average over all 48 quaternions or an extra quotient by their unit group.

Normalizing q to $q/\sqrt{5}$ does not change r_q . Quaternionic conjugation identifies the imaginary unit sphere with $\text{SO}(3)/\text{SO}(2)$, and q and $-q$ give the same rotation. For $(1 + 2\mathbf{i})/\sqrt{5}$ direct multiplication gives

$$\mathbf{i} \mapsto \mathbf{i}, \quad \mathbf{j} \mapsto (-3\mathbf{j} + 4\mathbf{k})/5, \quad \mathbf{k} \mapsto (-4\mathbf{j} - 3\mathbf{k})/5.$$

These are the columns of R_x in (60); cyclic permutation gives R_y, R_z . Conjugating a quaternion changes the sign of its imaginary coordinate and gives its normalized inverse. Thus the six

representatives already include inverse rotations, with one copy each. Division of the sum in (LPS5) by six gives $2\sqrt{5}/6 = \sqrt{5}/3$, and squaring gives $1 - 5/9 = 4/9$.

For the full-group statement, the complexified spherical space is the orthogonal sum of the harmonic spaces $\mathcal{H}_\ell$, $\ell \geq 0$. These are precisely all irreducible unitary representations of $\mathrm{SO}(3)$, one copy of each. The degree-zero space is the constants. The regular representation on $\mathrm{SO}(3)$ contains the same irreducibles with multiplicity $2\ell + 1$. Convolution acts on each copy by the same matrix, so its mean-zero operator norm is the supremum of the same matrix norms on both spaces. No extra invariant subspace is removed: on the sphere the constant is the only $\mathrm{SO}(3)$ -invariant function. The symmetric alphabet makes the inverse/right versus left convention immaterial for these norms. Half-integer-spin representations occur on $\mathrm{SU}(2)$ but not on its $\mathrm{SO}(3)$ quotient; this argument makes no claim about those extra blocks. $\square$

Source boundary. The original LPS publisher records were checked, but their complete 1986/1987 page images were not obtained in this revision. We therefore do not invent an original theorem number or label the secondary restatement as an inspected original proof. Equation (LPS5) states exactly the arithmetic line required, and the proposition proves every subsequent convention change. The effective theorem retains this explicit external dependency. The main width classification is conditional on an arbitrary full gap and does not depend on this numerical example or on an original-page attribution.

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